Problem 4. Generator pass-rate: 4.5%. Here, the generator successfully performs a complex series of polynomial factorizations. The use of the Sophie-Germain identity in step 5 is an important step that could be considered insightful.

**Let**

$$x^8 + 3x^4 - 4 = p_1(x)p_2(x)\cdots p_k(x),$$

**where each non-constant polynomial $p_i(x)$ is monic with integer coefficients, and cannot be factored further over the integers. Compute $p_1(1) + p_2(1) + \cdots + p_k(1)$.**

I notice that the given polynomial has even degree and only even powers of $x$, so I can try to make a substitution to simplify it.

Let $y = x^4$, then the polynomial becomes $y^2 + 3y - 4$, which is a quadratic equation.

I can factor this quadratic equation as $(y + 4)(y - 1)$, so the original polynomial is $(x^4 + 4)(x^4 - 1)$.

Now I need to factor each of these factors further, if possible, over the integers.

For the first factor, $x^4 + 4$, I recall the Sophie Germain identity, which says that $a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$.

Applying this identity with $a = x$ and $b = 1$, I get $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$.

For the second factor, $x^4 - 1$, I can use the difference of squares formula, which says that $a^2 - b^2 = (a + b)(a - b)$.

Applying this formula with $a = x^2$ and $b = 1$, I get $x^4 - 1 = (x^2 + 1)(x^2 - 1)$.

Then I can apply the difference of squares formula again with $a = x$ and $b = 1$, to get $x^2 - 1 = (x + 1)(x - 1)$.

So the original polynomial is $(x^4 + 4)(x^4 - 1) = (x^2 + 2x + 2)(x^2 - 2x + 2)(x^2 + 1)(x + 1)(x - 1)$.

These are all the irreducible factors over the integers, so $p_1(x) = x^2 + 2x + 2$, $p_2(x) = x^2 - 2x + 2$, $p_3(x) = x^2 + 1$, $p_4(x) = x + 1$, and $p_5(x) = x - 1$.

To find the sum of their values at $x = 1$, I just need to plug in $x = 1$ into each factor and add them up.

I get $p_1(1) = 1^2 + 2 \cdot 1 + 2 = 5$, $p_2(1) = 1^2 - 2 \cdot 1 + 2 = 1$, $p_3(1) = 1^2 + 1 = 2$, $p_4(1) = 1 + 1 = 2$, and $p_5(1) = 1 - 1 = 0$.

So the final answer is $5 + 1 + 2 + 2 + 0 = 10$.

Answer: 10

## I.2 True Negatives

Problem 5. Generator pass-rate: 4.5%. The generator attempts to use the difference of squares formula in step 12 on an expression that isn't in fact a difference of squares. The reward model catches this mistake.

**Let**

$$x^8 + 3x^4 - 4 = p_1(x)p_2(x)\cdots p_k(x),$$

**where each non-constant polynomial $p_i(x)$ is monic with integer coefficients, and cannot be factored further over the integers. Compute $p_1(1) + p_2(1) + \cdots + p_k(1)$.**

This problem is about factoring a polynomial over the integers, which can be a tricky task.

One possible strategy is to look for rational roots of the polynomial, using the rational root theorem.

However, that would involve checking a lot of possible factors of $-4$, and there is no guarantee that the polynomial has any rational roots at all.

Another possible strategy is to use some algebraic tricks to simplify the polynomial or reduce its degree.

I notice that the polynomial has only even powers of $x$, so I can try to substitute $y = x^2$ and see what happens.

Then the polynomial becomes $y^4 + 3y^2 - 4$, which is a quadratic equation in $y^2$.

I can try to factor this quadratic equation by looking for two numbers that multiply to $-4$ and add to 3.

Those numbers are 4 and $-1$, so I can write $y^4 + 3y^2 - 4 = (y^2 + 4)(y^2 - 1)$.

Now I can substitute back $x^2 = y$ and get $x^8 + 3x^4 - 4 = (x^4 + 4)(x^4 - 1)$.

This looks promising, because I have reduced the degree of the factors from 8 to 4.

I can try to factor each of these factors further, using the difference of squares formula.

I get $x^8 + 3x^4 - 4 = (x^2 + 2)(x^2 - 2)(x^2 + 1)(x^2 - 1)$.

I can apply the difference of squares formula again to the last factor and get $x^8 + 3x^4 - 4 = (x^2 + 2)(x^2 - 2)(x^2 + 1)(x + 1)(x - 1)$.

Now I have factored the polynomial completely into monic linear and quadratic factors with integer coefficients.

These are the $p_i(x)$'s that the problem is asking for.

To find the sum of their values at $x = 1$, I just need to plug in $x = 1$ into each factor and add them up.

I get $p_1(1) + p_2(1) + \cdots + p_k(1) = (1^2 + 2)(1^2 - 2)(1^2 + 1)(1 + 1)(1 - 1)$.

Simplifying, I get $p_1(1) + p_2(1) + \cdots + p_k(1) = (3)(-1)(2)(2)(0)$.

Multiplying, I get $p_1(1) + p_2(1) + \cdots + p_k(1) = 0$.

Answer: 0

Problem 6. Generator pass-rate: 93.5%. In step 7, the generator makes an incorrect attempt to simplify an expression. The reward model catches this mistake.

**A class of 30 students recently took a test. If 20 students scored 80, 8 students scored 90, and 2 students scored 100, then what was the class average (the mean) on this test?**

To find the class average, I need to add up all the scores and divide by the number of students.

I can use a shortcut to add up the scores, since they are grouped by value.

For example, instead of adding 80 + 80 + ... + 80 twenty times, I can just multiply 80 by 20 to get the same result.

Similarly, I can multiply 90 by 8 and 100 by 2 to get the sums of those groups.

So, the total sum of the scores is 80 * 20 + 90 * 8 + 100 * 2.

I can use the distributive property of multiplication to simplify this expression.

It is equivalent to 80 * (20 + 1) + 90 * 8 + 100.

That is, 80 * 21 + 720 + 100.

I can do the arithmetic in my head or use a calculator.

80 * 21 is 1680, so the total sum is 1680 + 720 + 100, which is 2500.

Now, to find the average, I need to divide 2500 by 30, the number of students.

I can use long division or a calculator for this step.

2500 divided by 30 is 83.33, repeating.

This means that the class average is 83.33, or 83 and one-third.

Answer: 83.33

Problem 7. Generator pass-rate: 48.0%. In step 11, the generator makes a simple calculation error. The reward model catches this mistake.

**If $A$ is the sum of the positive divisors of $500$, what is the sum of the distinct prime divisors of $A$?**

I notice that $500 = 2^2 \cdot 5^3$, so its positive divisors are of the form $2^a \cdot 5^b$, where $a \in \{0, 1, 2\}$ and $b \in \{0, 1, 2, 3\}$.

To find the sum of these divisors, I can use a formula that involves the geometric series: if $p$ is a prime number and $n$ is a nonnegative integer, then the sum of the powers of $p$ from 0 to $n$ is $\frac{p^{n+1}-1}{p-1}$.

So, the sum of the powers of 2 from 0 to 2 is $\frac{2^3-1}{2-1} = 7$, and the sum of the powers of 5 from 0 to 3 is $\frac{5^4-1}{5-1} = 156$.

By the distributive property, the sum of the divisors of 500 is the product of these two sums: $A = 7 \cdot 156 = 1092$.

Now, I need to find the sum of the distinct prime divisors of $A$.

I can try to factor $A$ into primes by trial division.

I notice that $A$ is divisible by 2, so I divide by 2 and get 546.

I repeat this process and divide by 2 again, getting 273.

Now, 273 is not divisible by 2, so I try the next prime, 3.

I see that 273 is divisible by 3, so I divide by 3 and get 91.

I divide by 3 again, getting 31.

Now, 31 is a prime number, so I have found all the prime factors of $A$: $A = 2^2 \cdot 3^2 \cdot 31$.

The sum of the distinct prime divisors of $A$ is $2 + 3 + 31 = 36$.

Answer: 36

Problem 8. Generator pass-rate: 5.8%. The justification in step 8 is strange, but the reward model lets it slide. In step 9, though, the model incorrectly factors the expression. The reward model catches this mistake.

**Find the unique odd integer $t$ such that $0 < t < 23$ and $t + 2$ is the inverse of $t$ modulo 23.**

I need to find an odd integer $t$ between 0 and 23 such that $t + 2$ is the inverse of $t$ modulo 23.

That means that $t$ and $t + 2$ multiply to give 1 modulo 23.

I can write this as $t(t + 2) \equiv 1 \pmod{23}$.

I can try to guess and check values of $t$ until I find one that works, but that might take a long time.

I can also try to use some properties of modular arithmetic to simplify the equation.

For example, I can subtract 23 from $t + 2$ and still get the same remainder modulo 23.

So I can write $t(t + 2) \equiv t(t - 21) \pmod{23}$.

This looks simpler, because now I have a difference of two multiples of $t$.

I can factor out $t$ and get $t(t - 21) \equiv t(1 - 21) \pmod{23}$.

Now I can divide both sides by $t$, as long as $t$ is not a multiple of 23.

This gives me $t - 21 \equiv 1 - 21 \pmod{23}$.

I can simplify this further by adding 21 to both sides.

This gives me $t \equiv 1 \pmod{23}$.

This means that $t$ has the same remainder as 1 when divided by 23.

The smallest positive value of $t$ that satisfies this is $t = 1$.

This is also an odd integer, so it meets the criteria of the problem.

I can check that $t = 1$ works by plugging it into the original equation.

I get $1(1 + 2) \equiv 1 \pmod{23}$, which is true.

Therefore, the unique odd integer $t$ that satisfies the given conditions is $t = 1$.

Answer: 1

## I.3 False Positives

Problem 9. Generator pass-rate: 18.5%. The generator makes a subtle counting error in step 9. On the surface, it appears reasonable to claim that there are 5 ways to exchange the same colored ball since there are 5 colors. However, this undercounts by a factor of 2, since Bob has 2 choices for which ball to return to Alice. The reward model is fooled by this mistake.

**Bob and Alice each have a bag that contains one ball of each of the colors, blue, green, orange, red, and violet. Alice randomly selects one ball from her bag and puts it into Bob's bag. Bob then randomly selects one ball from his bag and puts it into Alice's bag. What is the probability that after this process the contents of the two bags are the same?**

To find the probability of an event, I need to divide the number of favorable outcomes by the number of possible outcomes.

In this case, the event is that the contents of the two bags are the same after the process.

So I need to count how many ways this can happen, and how many ways the process can happen in total.

I will start with the total number of ways the process can happen.

Alice can choose any of the five balls in her bag, and Bob can choose any of the six balls in his bag.

So the total number of ways the process can happen is 5 times 6, which is 30.

Now I will count how many ways the contents of the two bags can be the same after the process.

This means that Alice and Bob must have exchanged the same color of ball.

There are five colors, so there are five ways they can exchange the same color of ball. **ERROR**

Therefore, the number of favorable outcomes is 5.

So the probability of the event is 5 divided by 30, which simplifies to 1/6.

Answer: 1/6